

Predictive Modeling & Algorithm Selection

1. Objective

This week's assignment focuses on the predictive modeling phase. The goal is to simulate the process of selecting and justifying appropriate predictive algorithms that can forecast customer trends or behaviors in the beauty & wellness domain. The task requires connecting business insights with technical methods, ensuring that model choice aligns with the data characteristics and the problem defined earlier.

2. Expected Deliverables

- A DOC file outlining the modeling strategy and rationale for algorithm selection.
- Detailed description of at least two predictive models (e.g., Linear Regression, Decision Trees, Clustering).
- Discussion of evaluation metrics (RMSE, precision, recall, F1-score, accuracy).
- Predictions about model performance and potential business impact.
- Explanation of how model outcomes will be interpreted for actionable insights.

3. Key Steps

1. Review the Week 1 problem definition and align it with predictive tasks.
2. Research predictive models using Python libraries (scikit-learn, statsmodels).
3. Compare models, highlighting strengths and weaknesses in context.
4. Define validation methods (train/test split, cross-validation, bootstrapping).
5. Discuss pitfalls (overfitting, underfitting, data imbalance) and mitigation strategies.

4. Candidate Models

Model 1: Linear Regression

- **Use Case:** Forecasting continuous outcomes such as monthly sales or customer spend.
- **Strengths:** Simple, interpretable, effective for linear relationships.
- **Weaknesses:** Sensitive to outliers, poor fit for non-linear data.
- **Business Insight:** Coefficients show how factors like price, promotions, or seasonality influence demand.

Model 2: Decision Trees / Random Forest

- **Use Case:** Predicting categorical outcomes such as churn (yes/no) or product preference.
- **Strengths:** Handles non-linear relationships, interpretable splits, robust with ensembles.
- **Weaknesses:** Risk of overfitting if not pruned; Random Forest less interpretable.
- **Business Insight:** Tree splits reveal key drivers of customer behavior (e.g., promo responsiveness, region).

Model 3 (Optional): Clustering (KMeans)

- **Use Case:** Customer segmentation based on purchase frequency, spend, and sentiment.
- **Strengths:** Groups customers into actionable segments.

- **Weaknesses:** Requires careful choice of k ; sensitive to scaling.
- **Business Insight:** Segments guide targeted marketing and personalized offers.

5. Evaluation Metrics

- **Regression Models:** RMSE, MAE, R^2 .
- **Classification Models:** Accuracy, Precision, Recall, F1-score, ROC-AUC.
- **Clustering Models:** Silhouette score, Davies-Bouldin index.
- **Business Mapping:** Metrics tied to KPIs like retention rate, conversion uplift, and revenue growth.

6. Validation & Robustness

- **Train/Test Split:** 70/30 or 80/20.
- **Cross-Validation:** k -fold CV for stability.
- **Regularization:** Lasso/Ridge for regression.
- **Pruning/Ensembling:** To reduce overfitting in trees.
- **Resampling:** SMOTE or class weighting for imbalanced data.

7. Potential Pitfalls & Mitigation

- **Overfitting:** Use cross-validation, limit tree depth, apply regularization.
- **Underfitting:** Add interaction features, use more complex models.
- **Data Imbalance:** Apply resampling or weighted metrics.
- **Interpretability vs Accuracy:** Balance simple interpretable models with complex high-accuracy ones.

8. Workload Estimate (30–35 Hours)

- 6 hrs → Problem review + data alignment.
- 10 hrs → Model research + coding experiments.
- 8 hrs → Validation + evaluation metrics.
- 6–8 hrs → Documentation + business mapping.

9. DOC File Structure

Title Page Executive Summary 1. Introduction 2. Problem Alignment 3. Candidate Models 4. Evaluation Metrics 5. Validation & Robustness 6. Pitfalls & Mitigation 7. Business Insights Mapping 8. Implementation Plan & Timeline 9. Appendix (Python code snippets, plots, glossary)